The value at each location in the s-plane consists of a *real part* and an *imaginary part*. The real part is found by multiplying the time domain signal by the exponentially weighted cosine wave and then integrated from -∞ to ∞. The imaginary part is found in the same way, except the exponentially weighted sine wave is used instead. It looks like this in equation form, using the real part of A&A′ as an example:

$$ReX(\sigma=1.5,\omega=\pm 40) = \int_{-\infty}^{\infty} x(t)\cos(40t)\,e^{-1.5t}\,dt$$

Figure 32-3 shows an example of a time domain waveform, its frequency spectrum, and its s-domain representation. The example time domain signal is a rectangular pulse of width *two* and height *one*. As shown, the complex Fourier transform of this signal is a sinc function in the real part, and an entirely zero signal in the imaginary part. The s-domain is an undulating two-dimensional signal, displayed here as topographical surfaces of the real and imaginary parts. The mathematics works like this:

$$X(s) = \int_{-\infty}^{\infty} x(t)\,e^{-st}dt = \int_{-1}^{1} 1\,e^{-st}dt$$

In words, we start with the definition of the Laplace transform (Eq. 32-1), plug in the unity value for $x(t)$, and change the limits to match the length of the nonzero portion of the time domain signal. Evaluating this integral provides the s-domain signal, expressed in terms of the complex location, s, and the complex value, $X(s)$:

$$X(s) = \frac{e^{s} - e^{-s}}{s}$$

While this is the most compact form of the answer, the use of complex variables makes it difficult to understand, and impossible to generate a visual display, such as Fig. 32-3. The solution is to replace the complex variable, s, with $\sigma + j\omega$, and then separate the real and imaginary parts:

$$ReX(\sigma,\omega) = \frac{\sigma\cos(\omega)[e^{\sigma}-e^{-\sigma}] + \omega\sin(\omega)[e^{\sigma}+e^{-\sigma}]}{\sigma^2+\omega^2}$$

$$ImX(\sigma,\omega) = \frac{\sigma\sin(\omega)[e^{\sigma}+e^{-\sigma}] - \omega\cos(\omega)[e^{\sigma}-e^{-\sigma}]}{\sigma^2+\omega^2}$$